

Lab Report

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Experiment Name:

Measurement of serum protein content

Purpose of the experiment:

The total protein content of serum samples was measured by a colourimetric method using a biuret reagent, and the serum protein concentration was analysed for compliance with the normal reference range.

Experimental Materials and Methods

Experimental materials:

1. serum samples
2. Standard protein solution (concentration: 50 mg/mL)

3. 0.9% NaCl solution

4. biuret reagent

5. spectrophotometer

6. 37°C water bath

Experimental Methods:

1. Blank group: add 0.9% NaCl solution 3.0 mL and 3.0 mL of biuret reagent, mix well and leave to react.

2. Standard group: add 2.5 mL of standard protein solution and 3.0 mL of biuret reagent, mix well and leave to react. 3.

3. test group: add 2.95 mL of serum sample and 3.0 mL of biuret reagent, mix well and leave to react. 4. water bath heating: all serum samples should be heated in a water bath.

4. water bath heating: incubate all reaction mixes in a 37°C water bath for 10 minutes to ensure that the reaction develops its colour.

5. Absorbance measurement: Immediately after incubation, read the absorbance values of each group at 540 nm on a spectrophotometer.

Experimental data

groups	added ingredient	Water bath conditions	absorbance (A)	protein concentration (C)
Blank	0.9% NaCl 3.0 mL + biuret reagent 3.0 mL	37°C, 10 mins	0.000	0
Standard	standard protein 2.5 mL + biuret reagent 3.0 mL	37°C, 10 mins	0.193	50 mg/mL
Test	serum sample 2.95 mL + biuret reagent 3.0 mL	37°C, 10 mins	0.132	34.2 mg/mL

data processing

Formula:

$$C_{test} = \frac{A_{test}}{A_{standard}} \times C_{standard}$$

Substitute data:

$$C_{test} = \frac{0.132}{0.193} \times 50 = 34.2 \text{ mg/ml}$$

Results:

The serum protein concentration in the test group was 34.2 mg/ml.

Experimental results and analyses

1. Analysis of results:

the serum protein concentration in the test group was 34.2 mg/mL, which is lower than the reference value for normal human total serum protein (60-80 mg/mL).

The results were more reliable during the incubation step.

The incubation step resulted in more adequate conditions for the colour development reaction and more reliable results.

If the sample is not diluted and the serum protein concentration is low, it needs to be verified in conjunction with the source of the sample or further experiments.

2. possible causes:

Sample source:

Samples may be from pathological conditions (e.g., patients with impaired liver function, malnutrition, or chronic disease) where serum protein concentrations may be significantly lower than the normal range.

Experimental error:

The spectrophotometer was not calibrated accurately and may have affected the results.

Experimental manipulation problems:

Although the reaction conditions are optimised, the adequacy of mixing still requires attention.

Conclusion

The serum protein concentration of the test group was 34.2 mg/mL, which is lower than the normal reference range, as determined by the biuret reagent method under the condition of heating for 10 minutes at 37°C. The concentration of serum protein in the test group is lower than the normal reference range. Further analysis of the sample source is needed to rule out the possibility of experimental error.

Suggested experimental improvements

1. Optimise the standard curve:

Increase the concentration gradient (e.g., 10, 20, 30, 40, 50 mg/mL) and construct a complete standard curve to verify that the experimental data fall within the linear range of the standard curve.

2. Repeat testing of samples:

Repeat the absorbance values to ensure reproducibility and accuracy of the results.

Calibrate the spectrophotometer to confirm the proper condition of the equipment.

3. optimise operation:

Ensure that the reagents are well mixed with the sample, especially after the addition of biuret reagent, it should be mixed well to ensure adequate colour development reaction.

Notes:

1. Experimental Conditions:

In this experiment, neither the standard tube nor the test tube involved sample dilution, so no dilution factor was applied to the final concentration calculation.

2. Calculation with Dilution Factor:

If the sample is diluted during the experiment (e.g., diluted 10 times), the final concentration must be multiplied by the dilution factor to obtain the actual concentration. The formula is as follows:

$$C_{\text{actual}} = C_{\text{measured}} \times \text{Dilution Factor}$$

Where:

C_{measured} : Measured concentration without considering dilution

C_{actual} : Actual concentration after correcting for dilution

Example:

If the measured concentration is $C_{\text{measured}} = 10 \text{ mg/mL}$, and the sample was diluted 10 times, the actual concentration is:

$$C_{\text{actual}} = 10 \times 10 = 100 \text{ mg/mL}$$

Therefore, if the sample is diluted, the dilution factor must be included in the calculation to ensure the accuracy of the concentration results.

Answer to the questions

1. Common Methods for Determining Protein Concentration:

Several techniques are widely employed to measure protein concentration, each with its own set of advantages and disadvantages:

Bicinchoninic Acid (BCA) Assay:

Advantages:

Compatible with detergents and various chemicals.

Offers a broad dynamic range (20–2000 $\mu\text{g/mL}$).

Suitable for reasonably high-throughput analyses.

Disadvantages:

Requires the presence of aromatic amino acids.

Susceptible to interference from reducing agents and metal ions.

Inaccurate at concentrations below 2 $\mu\text{g/mL}$.

Results depend on the accuracy of protein standards.

Bradford Assay:

Advantages:

Quick and straightforward one-step procedure (less than 10 minutes).

Unaffected by reducing agents.

Disadvantages:

Necessitates the presence of aromatic amino acids.

Interference from common detergents and buffer components.

Results are contingent on the accuracy of protein standards.

UV Absorbance at 280 nm:

Advantages:

Rapid and simple.

No need for incubation or additional reagents.

Requires low sample volumes.

Disadvantages:

Depends on the presence of aromatic amino acids.

High susceptibility to interference from sample or buffer components.

Limited sensitivity.

Biuret Assay:

Advantages:

Applicable to a wide range of protein concentrations.

Relatively simple to perform.

Disadvantages:

Subjective colorimetric results.

Prone to interference from other substances in the sample.

Lowry Assay:

Advantages:

High sensitivity compared to some other methods.

Disadvantages:

Time-consuming with multiple reagent additions.

Sensitive to interfering substances.

2. Normal Range of Adult Serum Protein Concentration:

The typical reference range for total serum protein in adults is approximately 6.0 to 8.3 grams per deciliter (g/dL). This total includes:

Albumin: 3.5 to 5.0 g/dL.

Globulins: The remaining portion, with specific ranges varying among different types of globulins.

It's important to note that normal value ranges can vary slightly among different laboratories.

3. Applications of Spectrophotometry:

Spectrophotometry is widely used in various fields, such as:

Biological and Chemical Analysis: Measuring protein and nucleic acid concentrations, and enzyme activity.

Clinical Diagnostics: Detecting blood components like glucose, cholesterol, and haemoglobin.

Environmental Monitoring: Analyzing water quality and detecting pollutants.

Food and Beverage Industry: Checking nutrient content and colour consistency.

Pharmaceutical Industry: Identifying and testing drug purity.